

# ligase~ — Reference Manual

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Granular synthesizer / sampler / looper / delay for Pure Data

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# OVERVIEW

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ligase~ is a granular synthesizer/sampler/looper/delay with real-time recording, reel/splice management, spectral fog, filter, distortion and chaotic parameter modulation inspired by features of the Morphogene, Audiomulch's DLGranulator and Tidalcycles.

ligase~ implements asynchronous granular synthesis with splice-based sample organization. The engine operates on a fixed-size stereo buffer (10 minutes at the host sample rate) divided into segments called splices.

## Components

Reel: Circular stereo buffer ( $\text{MAX\_REEL\_SECONDS} * 48000$  samples)

Scheduler: Grain pool manager (configurable 1-2000 grains, default 200)

Recorder: Real-time input capture with sound-on-sound mixing

Envelope: Table-based amplitude shaping (parabolic, trapezoidal, cosine)

Grain pool allocation is configurable via ligase.conf file placed in the same directory as the external.

Format:  $\text{max\_grains} = N$  (range 1-2000).

## Grain Generation

1. Trigger timing controlled by IOT (interonset time): 0.0005-2.0 seconds
2. Grain size: 0.001-2.0 seconds (default 0.1)
3. Playhead position determines grain start point within current splice
4. Speed parameter controls playback rate and direction (-4.0 to 4.0)
5. Envelope applied via table lookup with linear interpolation
6. Maximum concurrent grains: 1 to pool\_size

## Grain Mixing

Constant-power stereo panning (0=left, 0.5=center, 1=right)

Per-grain amplitude scaling: 0.0-2.0 (default 1.0)

Summation of all active grains per sample

No automatic gain compensation

# INLETS

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1. Audio L - Left input + Messages + Bangs(metro/clock)
2. Audio R - Right input
3. GrainSize - Grain duration (0.001-10 seconds)
4. GrainStart - Normalized start position (0-1)
5. Speed - Playback rate (-4 to 4)
6. Organize - Splice selection (0-1)
7. ScanRate - Scan speed for mode 2 (-1000 to 1000)
8. SoS - Sound-on-sound mix (0-1)
9. IOT - Interonset time (0.0005-2 seconds)
10. MaxGrains - Max concurrent grains (1-pool\_size)
11. GDelay Time - Delay time (0-10 seconds)
12. GDelay Feedback - Feedback amount (0-1) (DD-4/Bencina only)
13. GDelay Tone - Lowpass filter (0-1) (DD-4/Bencina only)
14. GDelay Mix - Dry/wet mix (0-1)
15. Fog Mix - Spectral fog blend (0-1)
16. Moog Cutoff - Filter cutoff (20-20000 Hz)
17. Moog Resonance - Filter resonance (0-4)
18. Moog Mix - Filter dry/wet (0-1)
19. MIDI Pitch - MIDI note (1-127, 0=inactive)
20. Envelope Skew - Shape skew (0-1)
21. Amplitude - Grain amplitude (0-2)
22. Pan - Stereo position (0-1)

Unconnected inlets default to stored values (set via messages).

Zero values from unconnected inlets do not overwrite stored parameters.

## OUTLETS

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1. Signal - Audio output left
2. Signal - Audio output right
3. Bang - Splice end/wrap notification (modes 2 & 3)
4. Bang - Grain onset (every N grains, set via grain\_bang\_rate)

# MESSAGES

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## File Operations

load - Load a 32-bit float stereo WAV (any rate; plays at the host rate). Resolved relative to the patch directory. save - Save current reel to WAV

## Playback Control

play <0|1> - Start/stop playback record <0|1> - Start/stop recording recsplice - Record and create splice on stop recinput - Input-only recording, create splice on stop

## Recording Configuration

sos <0-1> - Sound-on-sound mix level sos\_mode <0|1> - 0=record only, 1=crossfade (default)

## Splice Navigation

shift - Move splice selection (integer) organize <0-1> - Select splice by normalized value splice - Add splice marker at current position

## Splice Management

clear\_splices - Remove all markers, clear buffer clear\_splices\_except\_current - Keep only current splice clear\_current\_splice - Remove current splice splice\_join\_right - Join current with right neighbor splice\_join\_all - Join all splices into one

## Splice Behavior

splice\_create\_pos <0|1|2> - 0=playback pos, 1=right, 2=end splice\_jump <0|1> - Jump to new splice on creation splice\_finish\_nav <0|1> - Finish playback before nav splice\_split <0|1> - Allow/prevent splitting current

## Playhead

playhead <1|2|3> - Static/Scanning/Clock Advance scanrate - Scan rate for mode 2 speed <-4 to 4> - Playback speed/direction clock\_advance\_quant <0|1> - Use current/quantized length clockstop - Stop clock, keep last BPM

## Grain Parameters

grainsize <0.001-10> - Grain duration in seconds grainstart <0-1> - Position within splice where grains begin (0=start, 1=end) iot <0.0005-2> - Interonset time in seconds maxgrains <1-pool\_size> - Max concurrent grains amplitude <0-2> - Grain amplitude pan <0-1> - Stereo position (0=left, 0.5=center, 1=right) pan\_mode <0|1> - Mono panning / Stereo balance saw\_cycles <0-64> - Saw envelope

modulation cycles saw\_depth <0-1> - Saw envelope modulation depth envelope <0|1|2|3|4> - Parabolic/Trapezoidal/Cosine/Gaussian/Exponential env\_skew <0-1> - Envelope shape skew grain\_bang\_rate - Bang outlet every N grains (0=off)

## Envelope

envelope <0|1|2> - Parabolic/Trapezoidal/Cosine env\_skew <0-1> - Envelope shape skew

## IOT Quantization

timesig / - Time signature quantize <1|2|4|8|16|32|64|128> - Note subdivision quant <0-1> - Quantization amount

## Grain Size Quantization

gs\_timesig / gs\_quantize gs\_quant <0-1>

## Delay Quantization

delay\_timesig / delay\_quantize delay\_quant <0-1>

## Grain Delay

gdelay\_time <0-9.5> - Delay time in seconds gdelay\_feed <0-1> - Feedback amount gdelay\_tone <0-1> - Lowpass filter (0=dark, 1=bright) gdelay\_mix <0-1> - Dry/wet mix gdelay\_clear - Clear delay buffer

## Distortion

distortion <0-1> - Drive intensity distortion\_enable <0|1> - Enable/bypass distortion\_oversampling <1|2|4|8> - Oversampling factor distortion\_position <0|1> - Per-grain/Post-mix dist\_waveshaper\_mode <0|1|2|3|4> - Tanh/Arctan/Asymmetric/Blend/Polynomial distortion\_pre\_hp\_freq <30-500> - Pre-HP frequency distortion\_pre\_hp\_mix <0-1> - Pre-HP mix distortion\_post\_lp\_freq <2400-10000> - Post-LP frequency distortion\_post\_lp\_mix <0-1> - Post-LP mix distortion\_notch\_freq - Notch center frequency distortion\_notch\_bw - Notch bandwidth distortion\_notch\_mix <0-1> - Notch mix dist\_emphasis\_mode <0|1> - HP/LP pre-emphasis dist\_emphasis\_freq <100-5000> - Emphasis frequency dist\_pregain <0.1-10> - Pre-saturation gain dist\_curve\_blend <0-1> - Tanh/arctan morph dist\_drive\_pos <1-20> - Positive drive (asymmetric) dist\_drive\_neg <1-20> - Negative drive (asymmetric) dist\_poly\_c1 <-10 to 10> - Linear coefficient dist\_poly\_c2 <-10 to 10> - Quadratic coefficient dist\_poly\_c3 <-10 to 10> - Cubic coefficient

## Fog (Spectral Effect)

fog\_smear\_bins <0-32> - Frequency smear width (bins) fog\_smear\_enable <0|1> - Enable/disable smear fog\_smear\_onset\_curve <0|1|2> - Linear/Exponential/Logarithmic fog\_smear\_onset\_amount <0-1> - Smear onset depth fog\_mag\_cutoff <0.1-20> - Magnitude filter cutoff (Hz) fog\_mag\_resonance <0.1-10> - Magnitude filter Q fog\_phase\_cutoff <0.1-20> - Phase filter cutoff (Hz) fog\_specmagfilter\_enable <0|1> - Enable/disable temporal filter fog\_specmagfilter\_onset\_curve <0|1|2>



- Linear/Exponential/Logarithmic fog\_specmagfilter\_onset\_amount <0-1> - Temporal filter onset depth fog\_stereo\_filter\_mode <0|1> - Shared/Independent stereo state fog\_position <0|1> - Per-grain/Post-mix mode

#### Moogladder Filter

moog\_cutoff <20-20000> - Cutoff frequency (Hz) moog\_resonance <0-4> - Resonance amount moog\_mix <0-1> - Dry/wet mix moog\_enable <0|1> - Enable/bypass

#### Delay Modes

delay\_mode <0|1|2> - DD-4/Bencina/Stut stut - Trigger stut effect stut\_reps <1-16> - Stut repetitions stut\_reduction <0-1> - Gain decay per repeat stut\_spacing - Spacing between repeats bencina\_iot - Bencina grain spacing bencina\_grainsize - Bencina grain size bencina\_wrap <0|1> - Global/Splice wrap mode bencina\_clear - Clear bencina grains

#### Sphere Simulation

sphere\_kick - Apply velocity impulse sphere\_damping <0-1> - Damping coefficient sphere\_elasticity <0-1> - Bounce elasticity sphere\_reset - Reset to initial state sphere\_mode <0-6> - Output mode (X/Y/Z/VelX/VelY/VelZ/VelMag)

#### Parameter Ranging

param\_range - Set parameter range rand\_type - Assign generator param\_lock ... - Disable modulation and lock parameter(s) at their current modulated value. param\_invert <0|1> - Invert a parameter's modulation output (mirror around its range). param\_slew <0-1> - Exponential smoothing on a parameter's modulation output. param\_base\_value - Set the PERLIN\_2D Y-coordinate (decorrelation) for a parameter.

#### Generators

rand\_1, rand\_2, rand\_3, rand\_4 perlin\_1d\_1, perlin\_1d\_2, perlin\_1d\_3, perlin\_1d\_4 perlin\_2d\_1, perlin\_2d\_2, perlin\_2d\_3, perlin\_2d\_4 lorenz\_1, lorenz\_2, lorenz\_3, lorenz\_4 nbody\_1, nbody\_2, nbody\_3, nbody\_4 sphere\_1, sphere\_2, sphere\_3, sphere\_4 saw\_1, saw\_2, saw\_3, saw\_4 sine\_1, sine\_2, sine\_3, sine\_4 square\_1, square\_2, square\_3, square\_4

#### Generator Modulation Outlet Send

The four send outlets (modout1–modout4) are configured like any modulatable parameter — by addressing the param name modoutN with the unified messages. There are no dedicated modout\_source/modout\_range commands: rand\_type modoutN - assign a generator to outlet N param\_range modoutN - set the outlet's output range

#### Noise Frequency

noise\_freq - Set all generator speeds noise\_freq\_1|2|3|4 - generator instance speeds

## N-Body Configuration

nbody\_mode - Set output mode (0-10) nbody\_epsilon <0.01-1> - Softening parameter nbody\_damping <0-0.1> - Energy damping nbody\_pump <0-0.01> - Energy injection nbody\_reset - Reset to initial conditions

## Sphere Configuration

sphere\_kick - Apply velocity impulse sphere\_damping <0-1> - Damping coefficient sphere\_elasticity <0-1> - Bounce elasticity sphere\_reset - Reset to initial state sphere\_mode <0-6> - Output: X/Y/Z/VelX/VelY/VelZ/VelMag

## Pitch Control

pitch\_mode <0|1|2|3|4> - Off/Semitones/Range/Scale/MIDI pitch\_semitones <-24 to 24> - Fixed transposition pitch\_range - Semitone range pitch\_rand\_type - Random generator for pitch pitch\_scale ... - Scale definition (semitones)

## Defaults

### Core Parameters

grainsize: 0.1 seconds grainstart: 0.5 (center of splice) speed: 1.0 (normal forward) iot: 0.1 seconds maxgrains: 4 amplitude: 1.0 pan: 0.5 (center) pan\_mode: 0 (constant-power mono panning) saw\_cycles: 0.0 (off) saw\_depth: 0.0 (off) env\_skew: 0.5 (symmetric)

### Envelope

type: ENVELOPE\_COSINE table\_length: 4096 samples

### Playhead

mode: PLAYHEAD\_MODE\_STATIC (mode 1) scanrate: 1.0

### Recording

sos: 0.5 (50/50 mix) sos\_mode: 1 (Morphagene-style crossfade) mode: RECORD\_MODE\_OVERDUB

### Splice

create\_pos: 0 (at playback position) jump: 0 (stay in current) finish\_nav: 0 (immediate) split: 0 (allow split)

### Effects

fog\_mix: 0.0 (dry) fog\_smear\_bins: 3 fog\_smear\_enable: 1 fog\_smear\_onset\_curve: 2 (logarithmic) fog\_smear\_onset\_amount: 0.8 fog\_mag\_cutoff: 2.5 Hz fog\_mag\_resonance: 0.5 fog\_phase\_cutoff: 3.0 Hz fog\_specmagfilter\_enable: 1 fog\_specmagfilter\_onset\_curve: 2 (logarithmic) fog\_specmagfilter\_onset\_amount: 1.0 fog\_stereo\_filter\_mode: 1 (independent) fog\_position: 1 (post-

mix) fog\_pool\_size: 4 (configurable via ligase.conf) gdelay\_time: 0.0 (off) gdelay\_feed: 0.0 gdelay\_tone: 0.5 gdelay\_mix: 0.0 delay\_mode: 0 (DD-4) delay\_quant\_note: 16 (1/16) delay\_quant\_amount: 0.0 (disabled) stut\_reps: 4 stut\_reduction: 0.5 stut\_spacing: 62.5 ms bencina\_iot: 50.0 ms bencina\_grainsize: 0.1 seconds bencina\_wrap: 1 (splice) distortion: 0.0 (clean) distortion\_enable: 1 distortion\_oversampling: 4x distortion\_position: 1 (post-mix) moog\_cutoff: ~1000 Hz moog\_resonance: 0.0 moog\_mix: 0.0 moog\_enable: 1

#### Distortion Filters

pre\_hp\_freq: 30 Hz pre\_hp\_mix: 0.5 post\_lp\_freq: 16000 Hz post\_lp\_mix: 0.5 notch\_freq: 3000 Hz notch\_bandwidth: 500 Hz notch\_mix: 0.0 (inactive) emphasis\_freq: 800 Hz emphasis\_mode: 0, EMPHASIS\_MODE\_HP pregain: 1.0

#### Pitch Control

mode: PITCH\_MODE\_OFF semitones: 0.0 midi\_note: 60 (C4)

#### Quantization

All quant\_amount: 0.0 (off) timesig: 4/4 subdivision: 4 (quarter note)

#### Noise Generators

All frequency\_scale: 1.0 (normal) All rand\_type: RAND\_TYPE\_RAND All rand\_instance: 0 (generator 1)

#### Pool

pool\_size: 200 grains (configurable via ligase.conf)

#### Buffer

Sample rate: follows the host (Pd) rate — the reel capacity and saved WAVs use it (44.1/48/96 kHz etc.)  
Max length: 600 seconds Channels: 2 (stereo)

# PLAYBACK CONTROL

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Transport

play Start/stop grain triggering. 0 stops, non-zero starts.

When started:

- Checks reel length > 0. Posts error and aborts if empty.
- Initializes playback\_position to current splice start.
- Sets is\_playing and is\_triggering flags.

When stopped:

- Clears is\_playing and is\_triggering.
- Active grains finish their envelopes.

Default: 0 (stopped)

record Overdub into the CURRENT splice — Time Lag Accumulation. Argument: 0 stops, non-zero starts.

The record head loops within the current splice bounds, re-recording what is heard (the granular playback of the splice, mixed with the live input) back into the same splice. Because the splice being recorded is also the one being granulated, successive passes accumulate — layers build and any pitch transposition compounds across loops. SOS sets the input-vs-feedback balance (the feedback amount); the feedback is held just below unity and soft-limited so it sustains/decays rather than running away. Cross-splice recording is done by shifting the current splice (which moves the bounds) — overdub never extends the reel.

- Sets RECORD\_MODE\_OVERDUB; record head wraps within the current splice
- Stop: posts “recording stopped” (no new splice created)

recsplice Record a NEW splice of what is heard.

- Sets RECORD\_MODE\_NEW\_SPLICE
- Records the SOS-mixed monitor signal (live input crossfaded with the granular playback, per SOS) — i.e. “what you hear”. SOS acts as a VCA on the recorded signal.
- Posts starting position (current reel length); creates the splice when recording stops

recinput Record the live INPUT only — the one mode where SOS does NOT act as a VCA.

- Sets RECORD\_MODE\_INPUT\_ONLY

- Captures the raw input directly (no SOS, no feedback); ignores the sos parameter
- Creates a splice when stopped

## Playhead Modes

playhead 1|2|3 Set playhead behavior.

### Mode 1: Static (default)

Playback position fixed. grain\_start parameter (inlet 4) sets grain spawn location directly Grain position:  
 $\text{splice\_start} + (\text{grain\_start} \times \text{splice\_length})$

Detects grain\_start parameter wraps (change > 0.5), bangs outlet 3. No playback\_position advancement.

### Mode 2: Scanning

Playback position advances continuously at scan\_rate (samples per sample).

Grain position:  $\text{playback\_position} + (\text{grain\_start} \times \text{splice\_length})$

Advancement:  $\text{playback\_position} += \text{scan\_rate}$  every sample

Wraps at splice boundaries, bangs outlet 3 on wrap. Checks pending splice navigation after wrap.

Negative scan\_rate = reverse.

### Mode 3: Clock Advance

Playback position advances only when clock bang received (inlet 1).

Advance amount:  $\text{grain length} \times \text{sample\_rate}$

Grain length source: current or quantized (see clock\_advance\_quant)

Grain position:  $\text{playback\_position} + (\text{grain\_start} \times \text{splice\_length})$  between bangs Wraps, bangs outlet, checks pending navigation like Mode 2.

scanrate Set scanning speed for Mode 2.

1.0 = real-time forward

-1.0 = real-time reverse

0.0 = static

Default: 1.0

clock\_advance\_quant 0|1 Select grain length sourcefor Mode 3 advance.

0 = current actual grain length (varies per grain)

1 = quantized grid length (requires valid BPM and `gs_quant_grid_ms > 0`)

Default: 0

## Splice Structure

- Array of position markers (0-300 maximum)
- Each splice defined by start marker, extends to next marker or buffer end
- Current splice selected via `organize` parameter (0.0-1.0 maps to splice index)
- Shift message: relative navigation (shift 1 = next, shift -1 = previous)

## Splice Bounds

Grains read exclusively within current splice boundaries. Position wraps at `splice_end` back to `splice_start`. This creates independent loop regions within the buffer.

## Splice Navigation

`organize` Navigate to splice by `normalizedposition` (0.0-1.0).

Maps to index:  $\text{int}(\text{value} \times (\text{splice\_count} - 1))$

Clamps input to [0.0, 1.0]

If `splice_finish_nav = 0`: immediate navigation, resets `playback_position` to splice start

If `splice_finish_nav = 1`: queues navigation, executes after current splice wraps

shift Move splice selection by delta.

Adds delta to `current_splice` index.

Clamps to [0, count-1]. No wraparound.

Examples: shift 1 = next, shift -1 = previous

If `splice_finish_nav = 1`: queues as pending

If `splice_finish_nav = 0`: immediate

## Splice Creation

`splice` Create splice marker.

Position determined by `splice_create_pos`:

- 0: at `playback_position` (if 0 and reel exists, uses `reel_length`)

- 1: at current splice end boundary
- 2: at reel end

Rate limited: minimum 10ms between creation

Posts error if MAX\_SPLICES (64) reached

splice\_create\_pos 0|1|2 Set splice creation positionmode.

0 = at playback position (default)

1 = right of current splice boundary

2 = at end of reel (Morphogene append)

splice\_jump 0|1 Control navigation after creating new splice.

0 = stay in current splice (default)

1 = jump to newly created splice immediately

splice\_finish\_nav 0|1 Control navigation timing for organize and shift messages.

0 = immediate navigation (default)

1 = queue navigation, execute after current splice wraps

splice\_split 0|1 Control splice splitting behavior.

0 = allow splitting current splice (default)

1 = preserve current splice length

# RECORDING CONFIGURATION

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## Record Modes

Three recording modes, set by message commands. Mode determines recording behavior and splice creation.

record Overdub mode. Argument: 0 stops, non-zero starts.

- Sets RECORD\_MODE\_OVERDUB
- Start: positions record\_position at current\_splice\_start
- Records into current splice bounds
- Continues past splice boundary into adjacent splice (cross-splice recording)
- Mixing controlled by crossfade\_mix (sos value)
- Stop: Posts “recording stopped” only

recsplice New splice mode.

- Sets RECORD\_MODE\_NEW\_SPLICE
- Start: positions record\_position at reel length (append)
- Stores new\_splice\_start = current reel length
- Mixing controlled by crossfade\_mix (sos value)
- Stop: Creates splice marker at new\_splice\_start position, sets as current splice
- Posts splice creation confirmation with position

recinput Input-only mode.

- Sets RECORD\_MODE\_INPUT\_ONLY
- Start: positions record\_position at reel length (append)
- Stores new\_splice\_start = current reel length
- Records input directly, ignores sos/crossfade\_mix completely
- Stop: Creates splice marker at new\_splice\_start position, sets as current splice



- Posts splice creation confirmation with position

Default mode: RECORD\_MODE\_OVERDUB

SOS Mode Control

sos\_mode 0|1 Set Sound-on-Sound operational mode.

Mode 0: Record Only

- Output: Constant-power crossfade between input and granular
  - sos = 0.0: 100% granular (input\_gain = 0.0, granular\_gain = 1.0)
  - sos = 0.5: Equal power mix (both  $\approx 0.707$ )
  - sos = 1.0: 100% input (input\_gain = 1.0, granular\_gain = 0.0)
  - Formula:  $\text{out} = \text{input} \times \sin(\text{sos} \times \pi/2) + \text{granular} \times \cos(\text{sos} \times \pi/2)$
- Recording: happens before output mixing
  - Records raw input only (not the mixed output)
  - crossfade\_mix forced to 1.0 (full replacement, no sound-on-sound)
  - sos parameter does NOT affect recording in this mode
  - Always replaces buffer content with input signal
- Output and recording are decoupled:
  - Recording: Raw input  $\rightarrow$  buffer (replace mode)
  - Output: Input + granular constant-power mix (monitoring)

Mode 1: Morphogene (default)

- Output: Constant-power crossfade between input and granular
  - $\text{output} = \text{input} \times \sin(\text{sos} \times \pi/2) + \text{granular} \times \cos(\text{sos} \times \pi/2)$
  - sos = 0.0: 100% granular (input\_gain = 0.0, granular\_gain = 1.0)
  - sos = 0.5: Equal power mix (both  $\approx 0.707$ )
  - sos = 1.0: 100% input (input\_gain = 1.0, granular\_gain = 0.0)
  - sos from signal inlet (inlet 8) or stored sos\_value
  - Uses inlet value if non-zero, else uses stored value

- Clamped to [0.0, 1.0]
- NaN/Inf protection via isfinite() check
- Recording: what gets written depends on the record mode (see RECORDING CONFIGURATION):
  - recinput (INPUT\_ONLY): the raw live input — SOS bypassed (the one non-VCA mode)
  - recsplice (NEW\_SPLICE): the monitored output above — “what you hear” (the SOS crossfade of input and granular) — written to a new splice
  - record (OVERDUB): Time Lag Accumulation into the current splice. The monitored output is fed back into the splice each pass (SOS sets the input-vs-feedback balance), so layers and pitch accumulate; held stable by a sub-unity feedback coefficient and a tanh soft-limiter
- When no playback (reel length = 0): the monitored output is input × sos (linear VCA). recsplice/overdub capture that; recinput captures the raw input.

Default: 1 (Morphogene mode)

SOS Crossfade Parameter

sos Set crossfade mix value.

Range:

0.0 to 1.0 (clamped in processing) Default: 0.5

# SPLICE NAVIGATION

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## Direct Navigation

shift Navigate by relative offset.

Adds delta to current\_splice index. Clamps to [0, count-1]. No wraparound at boundaries.

- Positive: move right through array
- Negative: move left through array
- 0 stops at first splice
- count-1 stops at last splice

If finish\_before\_nav = 1: calculates target with wraparound  $((\text{target} \% \text{count}) + \text{count}) \% \text{count}$ , stores as pending, executes after playback wraps at splice boundary.

If finish\_before\_nav = 0: immediate navigation.

organize Navigate by normalized position.

Maps input [0.0-1.0] to discrete splice index:  $\text{int}(\text{value} \times (\text{count} - 1))$

Clamps input to [0.0, 1.0] before mapping.

Examples: 0.0 = first splice, 0.5 = middle splice, 1.0 = last splice

If finish\_before\_nav = 1: calculates target, stores as pending.

If finish\_before\_nav = 0: immediate navigation.

## Splice Creation

splice Create splice marker.

Position determined by create\_position setting:

Mode 0 (default): At playback\_position - If playback\_position = 0 and reel exists, uses reel\_length instead - If split\_current = 1 and position would split current splice (between splice\_start and

splice\_end), moves to splice\_end

Mode 1: At current splice end boundary (splice\_end) Mode 2: At reel end (reel\_length) Rate limited: minimum 10ms between creation. Posts error if violated.

Maximum: 64 splices (MAX\_SPLICES). Posts error if exceeded.

After creation, if jump\_to\_new = 1: sets current\_splice to new index, resets playback\_position to new splice start.

splice\_create\_pos 0|1|2 Set creation position mode. 0 = at playback position

1 = right of current splice

2 = at end of reel

Default: 0

splice\_jump 0|1 Control navigation after splice creation.

0 = stay in current splice (default)

1 = jump to newly created splice, reset playback\_position

splice\_finish\_nav 0|1 Control navigation timing for shift and organize.

0 = immediate navigation (default)

1 = queue navigation, execute when playback wraps at splice boundary

Pending navigation stored in pending\_splice. Cleared after execution.

splice\_split 0|1 Control current splice splitting.

0 = allow splitting current splice (default)

1 = preserve current splice length

When enabled (mode 1), if create\_position = 0 and calculated position falls within current splice bounds, position moves to splice\_end.

## Splice Removal

clear\_splices Remove all splice markers, clear entire audio buffer.

Calls splice\_clear() and reel\_clear(). Destructive operation. Sets count = 0, current\_splice = 0.

clear\_splices\_except\_current Remove all splices except current, clear buffer

except current splice audio.

Destructive. Extracts current splice bounds, clears buffer outside bounds, reorganizes markers:

- If current has right neighbor: creates two markers (position 0 and splice\_length)
- If current extends to reel end: creates one marker (position 0)

Current splice moves to position 0. Sets `current_splice = 0`.

`clear_current_splice` Remove current splice marker immediately.

Uses `splice_remove()` to delete marker at `current_splice` index. Shifts subsequent markers down. Active grains finish naturally.

Adjusts `current_splice` index if needed:

- If count becomes 0: `current_splice = 0`
- If `current_splice >= count`: `current_splice = count - 1`

Posts count of remaining splices.

Splice Joining

`splice_join_right` Join current splice with right neighbor.

Calculates right neighbor with wraparound:  $(\text{current\_splice} + 1) \% \text{count}$

Removes marker at right neighbor (boundary between them). Updates `current_splice` index if `right_splice <= current_splice`.

No effect if only one splice exists.

`splice_join_all` Join all splices into one continuous splice. Calls `splice_clear()`. Entire reel becomes single splice with no markers. Sets `count = 0`, `current_splice = 0`.

Posts number of splices joined.

# PLAYHEAD MODES

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Three modes control how playback position advances and where grains spawn within the current splice.

playhead 1|2|3 Set playhead mode.

Mode 1: Static (default)

Playback position fixed. grain\_start signal (inlet 4) directly controls grain spawn location.

Grain position:  $\text{splice\_start} + (\text{grain\_start} \times \text{splice\_length})$

grain\_start range: 0.0 to 1.0

0.0 = splice start, 0.5 = splice middle, 1.0 = splice end

Wrap detection: When grain\_start change exceeds 0.5 (indicating wrap from 1.0→0.0 or 0.0→1.0), bangs outlet 3.

playback\_position: Never advances. Remains at value set by play message (splice start).

grain\_start tracking: Stores prev\_grain\_start each block for wrap detection.

Mode 2: Scanning

Playback position advances continuously. grain\_start offsets grain spawn relative to moving playhead.

Grain position:  $\text{playback\_position} + (\text{grain\_start} \times \text{splice\_length})$  Wraps to stay within splice bounds.

Advancement:  $\text{playback\_position} += \text{scan\_rate}$  every audio sample

scan\_rate units: samples per sample

1.0 = real-time forward

1.0- = real-time reverse

0.0 = static (no advancement)

2.0 = double-speed forward

Boundary wrapping:

When  $\text{playback\_position} \geq \text{splice\_end}$ : wraps by subtracting splice\_length

When  $\text{playback\_position} < \text{splice\_start}$ : wraps by adding splice\_length

Bangs outlet 3 on wrap. Checks pending\_splice navigation after wrap, executes if queued.

### Mode 3: Clock Advance

Playback position advances only when clock bang received (inlet 1). Grains spawn at static position between bangs.

Grain position:  $\text{playback\_position} + (\text{grain\_start} \times \text{splice\_length})$  Wraps to stay within splice bounds.

Advancement trigger: Bang to inlet 1 sets clock\_bang\_received flag.

Advance amount:  $\text{grain\_length} \times \text{sample\_rate}$  (converted from seconds to samples)

Grain length source:

Determined by clock\_advance\_quant setting and current state.

If clock\_advance\_quant = 0: uses quantized\_grain\_size (current grain length from perform cycle)

If clock\_advance\_quant = 1 AND gs\_quant\_grid\_ms > 0 AND bpm > 1.0: Uses  $\text{gs\_quant\_grid\_ms} / 1000.0$  (quantized grid length in seconds)

Otherwise: uses quantized\_grain\_size

Per-bang behavior:

Flag checked once per perform cycle. If set, advances playback\_position, wraps at boundaries, bangs outlet 3 if wrapped, checks pending navigation, then clears flag.

Grains trigger normally at IOT rate at current playback\_position while flag is clear.

### Control Messages

scanrate Set scan rate for Mode 2.

Applied per audio sample as:  $\text{playback\_position} += \text{scan\_rate}$

Default: 1.0

clock\_advance\_quant 0|1 Select grain length source for Mode 3 advancement.

0 = use current grain length (varies per grain)

1 = use quantized grid length (requires valid BPM and gs\_quant\_grid\_ms)

Default: 0

# GRAIN PARAMETERS

---

## Interonset Time

iot Set time interval between grain starts.

Range: 0.0005 to 2.0 seconds

Clamped to bounds.

Converted to grain\_trigger\_period:  $\text{int}(\text{iot} \times \text{sample\_rate})$  samples

When  $\text{iot} < \text{grain\_size}$ : grains overlap

When  $\text{iot} > \text{grain\_size}$ : grains separate

When  $\text{iot} = \text{grain\_size}$ : continuous stream

Signal inlet 8: updates if value in range [0.001, 2.0], else keeps current value.

Quantization: If quant\_amount > 0 and BPM valid:

$\text{grain\_trigger\_period} = \text{base\_period} \times (1 - \text{quant\_amount}) + \text{quant\_period} \times \text{quant\_amount}$

Minimum grain\_trigger\_period enforced: 1 sample (prevents CPU overload)

Default: 0.1 seconds (100ms)

## Grain Size

grainsize Set grain duration.

Range: 0.001 to 10.0 seconds

Clamped to bounds.

Signal inlet 2: updates if value in range [0.001, 10.0], else keeps current value.

Default: 0.1 seconds (100ms)

## Grain Start Position

grainstart Set normalized grain spawn position within splice.

Range: 0.0 to 1.0

0.0 = splice start, 0.5 = splice middle, 1.0 = splice end



Signal inlet 4: updates if value in range [0.0, 1.0]

Special handling: if inlet reads 0.0 and stored value is non-zero, assumes inlet unconnected and preserves stored value.

Interacts with playhead mode:

- Mode 1 (Static):  $\text{grain\_pos} = \text{splice\_start} + (\text{grainstart} \times \text{splice\_length})$
- Mode 2 (Scanning):  $\text{grain\_pos} = \text{playback\_position} + (\text{grainstart} \times \text{splice\_length})$
- Mode 3 (Clock):  $\text{grain\_pos} = \text{playback\_position} + (\text{grainstart} \times \text{splice\_length})$

Default: 0.5 (middle)

Maximum Grains

maxgrains Set maximum concurrent active grains. Range: 1 to pool\_size

pool\_size read from ligase.conf (default: 200)

Clamped to [1, pool\_size]

Controls polyphony limit. Grain scheduler uses free-list allocation. When limit reached, oldest grains replaced.

Signal inlet 9: updates if value in range [1, pool\_size], else keeps current value.

Default: 4

Amplitude

amplitude Set grain output level.

Range: 0.0 to 2.0

Clamped to bounds.

Applied during grain triggering. Passed to scheduler\_trigger\_grain().

Signal inlet 19: amplitude\_in (read from perform cycle)

Default: 0.75

Per-grain amplitude scaling and randomization

There are two operational modes based on param\_range configuration:

Mode 1: Direct Control (param\_range disabled, default)

- Uses amplitude value directly from inlet or message

- No per-grain randomization
- Consistent amplitude across all grains

Mode 2: Modulated Range (param\_range enabled)

- Samples amplitude from [min, max] range per grain
- Uses inlet/message value as base for sampling
- Random source determines variation pattern
- Examples:

param\_range amplitude 0.5 1.0 → grains vary 0.5-1.0

rand\_type perlin\_1d\_1 amplitude → smooth amplitude evolution

rand\_type rand\_1 amplitude → random per-grain levels

Pan Mode

pan\_mode Select panning algorithm.

Mode 0: Constant-Power Mono Panning (default)

Sums stereo source to mono before panning

Each grain becomes a point source in the stereo field

Ideal for spatial positioning and grain cloud diffusion

Mode 1: Stereo Balance

Preserves stereo width of source material

Applies constant-power gains independently to L/R channels

Maintains stereo image character while adjusting balance

Ideal for stereo sources where width should be preserved

Default: 0 (constant-power mono panning)

Pan Law

Both modes use constant-power panning to maintain consistent perceived loudness:

$\text{pan\_angle} = \text{pan} * (\pi/2)$

$\text{left\_gain} = \cos(\text{pan\_angle}) // 1.0 \rightarrow 0.707 \rightarrow 0.0$

$\text{right\_gain} = \sin(\text{pan\_angle}) // 0.0 \rightarrow 0.707 \rightarrow 1.0$

This ensures  $\cos^2(\theta) + \sin^2(\theta) = 1$  at all pan positions, preventing the “hole in the middle” effect where center-panned signals sound quieter.

At center (pan=0.5): Both channels at 0.707 (-3dB) = 0dB total power

Pan Modulation

Pan can be modulated per-grain using param\_range:

Direct Control (param\_range disabled)

Uses pan value directly from inlet or message

No per-grain randomization

All grains at same stereo position

Default behavior

Modulated Range (param\_range enabled):

Samples pan position from [min, max] range per grain

Uses inlet/message value as base for sampling

Random source determines spatial pattern

Creates spatial diffusion effects

Examples:

param\_range pan 0.0 1.0

rand\_type perlin\_1d\_1 pan ← smooth stereo movement

param\_range pan 0.3 0.7

rand\_type rand\_1 pan ← random center-focused placement

param\_range pan 0.0 0.5

rand\_type lorenz\_1 pan ← chaotic left-side positioning

Spatial Cloud Effects:

Wide range + fast modulation = diffuse stereo cloud

Narrow range + slow modulation = subtle stereo shimmer

Per-grain randomization creates spatial depth impossible with traditional panning

## Envelope Shape

envelope 0|1|2 Set grain envelope type.

Type 0: Parabolic (Welch window)

Formula:  $1 - 4(x - 0.5)^2$

Smooth, gentle envelope. Quadratic curve.

Type 1: Trapezoidal

Linear attack (10% default), sustain, linear release (10% default). Hard attack/release with flat sustain. Percentages adjusted by skew.

Type 2: Cosine (Hann window, default)

Formula:  $0.5 \times (1 - \cos(2\pi \times x))$

Smoothest envelope. Reduces spectral artifacts.

Envelope table length: 4096 samples

Preserves skew value when type changes.

Default: ENVELOPE\_COSINE (type 2)

## Envelope Skew

Skew remaps time axis before envelope calculation, shifting peak position while maintaining smoothness.

env\_skew Set envelope attack/decay balance.

Range: 0.0 to 1.0

Clamped to bounds.

0.5 = symmetric (peak at center)

< 0.5 = shorter attack, longer decay (peak shifts left)

0.5 = longer attack, shorter decay (peak shifts right)

Regenerates envelope table immediately on change.

Signal inlet 18: updates if value in range [0.001, 1.0] and different from current, else keeps current value.

## Saw Envelope Modulation

Per-grain sawtooth wave carved into the envelope. Creates jagged, rhythmic amplitude spikes within each grain. The saw wave modulates the envelope shape: at depth=0 the envelope is unmodified, at depth=1 the saw wave fully carves the envelope into sharp peaks.

Formula:  $\text{env\_val} = \text{base\_env} * (1.0 - \text{depth} * (1.0 - \text{saw\_val}))$

saw\_cycles Number of sawtooth cycles per grain duration.

Range: 0.0 to 64.0. Default: 0.0 (off).

0.0 = no modulation. 1.0 = one ramp per grain. Higher values = more spikes. Non-integer values produce partial final cycles.

Signal inlet 21 or message. Supports param\_range modulation.

saw\_depth Modulation intensity.

Range: 0.0 to 1.0. Default: 0.0 (off).

0.0 = pure base envelope. 1.0 = maximum jaggedness (spikes drop to zero between peaks).

Signal inlet 22 or message. Supports param\_range modulation.

Grain Bang Output

grain\_bang\_rate Set grain onset bang rate.

Range: 0 to 1000

0 = disabled (no bangs)

1 = bang every grain

2 = bang every 2nd grain

N = bang every Nth grain

Bangs outlet 4 when grain\_bang\_counter reaches grain\_bang\_rate. Counter resets on rate change.

# TIMING & QUANTIZATION

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## Clock Management

Clock bangs to inlet 1 establish tempo. BPM calculated from the interval between bangs.

Bang to inlet 1: Calculate BPM and update quantization grids.

First bang: stores timestamp, no BPM calculation

Second bang onward: calculates interval, derives BPM

BPM calculation:  $60000.0 / \text{interval\_ms}$

Sets clock\_running flag. Recalculates all active quantization grids (IOT, grain size, delay) using new BPM.

In playhead Mode 3: sets clock\_bang\_received flag for playhead advance.

clockstop Clears clock\_running flag.

Preserves last calculated BPM and quantization grids for continued use. Posts current BPM value.

## IOT Quantization

Snaps interonset time to rhythmic grid.

timesig Set time signature for IOT quantization.

Format: “numerator/denominator” (e.g., “7/5”, “4/4”) Requires symbol argument. Parses using sscanf().

Validation: numerator > 0, denominator > 0

Posts error if invalid format or values.

Stores: time\_sig\_numerator, time\_sig\_denominator

Default: 4/4

quantize Set IOT quantization note subdivision. Valid values: 1, 2, 4, 8, 16, 32, 64, 128

Represents note division (16 = 1/16 note)

Stores quant\_note. Recalculates quant\_grid\_ms if BPM > 0:

Grid calculation:

$ms\_per\_whole\_note = (60000.0 / bpm) \times 4.0$

$quant\_grid\_ms = ms\_per\_whole\_note / quant\_note$

Posts grid size in milliseconds if BPM available, otherwise defers calculation.

Default: 16 (1/16 note)

quant Set IOT quantization amount (blend).

Range: 0.0 to 1.0

0.0 = no quantization (free timing)

1.0 = full quantization (locked to grid)

0.5 = 50% blend

Applied in perform cycle:

$grain\_trigger\_period = base\_period \times (1 - quant\_amount) + quant\_period \times quant\_amount$

Requires:  $quant\_amount > 0$ ,  $bpm > 1.0$ ,  $quant\_grid\_ms > 0$

Default: 0.0 (no quantization)

Grain Size Quantization

Snaps grain duration to rhythmic grid.

gs\_timesig Set time signature for grain size quantization.

Format: "numerator/denominator"

Validation and parsing identical to timesig.

Stores:  $gs\_time\_sig\_numerator$ ,  $gs\_time\_sig\_denominator$

Default: 4/4

gs\_quantize Set grain size quantization notesubdivision. Valid values: 1, 2, 4, 8, 16, 32, 64, 128

Calculation identical to quantize.

Stores  $gs\_quant\_note$ . Recalculates  $gs\_quant\_grid\_ms$  if  $BPM > 0$ .

Default: 16 (1/16 note)

gs\_quant Set grain size quantization amount.

Range: 0.0 to 1.0

Blend between free grain\_size and quantized grid.

Applied in perform cycle:

$\text{quant\_grain\_size\_sec} = \text{gs\_quant\_grid\_ms} / 1000.0$

$\text{quantized\_grain\_size} = \text{grain\_size} \times (1 - \text{gs\_quant\_amount}) + \text{quant\_grain\_size\_sec} \times \text{gs\_quant\_amount}$

Result assigned to scheduler->grain\_size.

Requires:  $\text{gs\_quant\_amount} > 0$ ,  $\text{bpm} > 1.0$ ,  $\text{gs\_quant\_grid\_ms} > 0$

Default: 0.0 (no quantization)

Delay Time Quantization

Snaps grain delay time to rhythmic grid.

delay\_timesig Set time signature for delayquantization.

Format: "numerator/denominator"

Validation and parsing identical to timesig.

Stores: delay\_time\_sig\_numerator, delay\_time\_sig\_denominator

Default: 4/4

delay\_quantize Set delay quantization note subdivision. Valid values: 1, 2, 4, 8, 16, 32, 64, 128

Calculation identical to quantize.

Stores delay\_quant\_note. Recalculates delay\_quant\_grid\_ms if BPM > 0.

Default: 16 (1/16 note)

delay\_quant Set delay quantization amount.

Range: 0.0 to 1.0

Blend between free delay\_time and quantized grid.

Applied in perform cycle:

$\text{quant\_delay\_sec} = \text{delay\_quant\_grid\_ms} / 1000.0$

$\text{quantized\_delay} = \text{current\_delay} \times (1 - \text{delay\_quant\_amount}) + \text{quant\_delay\_sec} \times \text{delay\_quant\_amount}$

Result assigned to grain\_delay->delay\_time.

Requires:  $\text{delay\_quant\_amount} > 0$ ,  $\text{bpm} > 1.0$ ,  $\text{delay\_quant\_grid\_ms} > 0$



Default: 0.0 (no quantization)

# PITCH & SPEED

---

pitch\_mode 0|1|2|3|4 Five pitch modes control relationship between speed parameter and playback rate.

Mode 0 - OFF (default)

speed

Speed parameter = playback rate directly.

speed\_range active if enabled.

Formula:  $\text{final\_speed} = \text{sample\_param\_range}(\text{speed\_range}, \text{speed\_inlet})$

Mode 1 - SEMITONES

Speed parameter = base pitch.

Fixed semitone transposition applied.

speed\_range bypassed.

Formula:  $\text{final\_speed} = \text{speed} * 2^{(\text{semitones}/12)}$

pitch\_semitones <-24 to 24>

Example: pitch\_semitones 7 = perfect fifth up

Mode 2 - RANGE

Speed parameter = base pitch.

Random semitone shift from range.

speed\_range bypassed.

Formula:  $\text{final\_speed} = \text{speed} * 2^{(\text{random\_semitones}/12)}$

Messages, see PARAMETER RANGES & MODULATION for generators:

pitch\_range

pitch\_rand\_type

Example:

pitch\_range -12 12 ( $\pm 1$  octave)

Mode 3 - SCALE

Speed parameter = base pitch.

Random note from scale/list of semitones (quantized pitch).

speed\_range bypassed.

Formula:  $\text{final\_speed} = \text{speed} * 2^{(\text{scale\_note}/12)}$

pitch\_scale ...

Example:

pitch\_scale 0 2 4 5 7 9 11 (major scale)

Up to 128 notes per scale.

Mode 4 - MIDI

Speed parameter = base pitch (assumed C4 = 60).

MIDI note input transposes from base.

speed\_range bypassed.

Formula:  $\text{final\_speed} = \text{speed} * 2^{((\text{midi\_note} - 60)/12)}$

MIDI input via inlet 19 (1-127, 0 = inactive)

Final speed clamped: -4.0 to 4.0 ( $\pm 48$  semitones / 4 octaves)

MIDI mode “latches” to the last valid note until pitch\_mode changes. Switching back to pitch\_mode 4 will resume with this latched note.

# DELAY

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Delay effect applied to the mixed grain output before recording. Three modes available: DD-4 (analog-style), Bencina (pitch-preserving granular), and Stut (quantized rhythmic). All modes share a 9.5-second stereo circular buffer (sized to the host rate) and common control parameters (time, feedback, tone, mix). Mode-specific parameters are set via messages.

## Parameters

`gdelay_time` Set delay time.

Range: 0.0 to 9.5 seconds

Clamped to bounds.

Smoothed internally to prevent clicks during time changes. Smoothing coefficient: 0.001 (approximately 20ms transition at 48kHz).

Signal inlet 10: updates if value in range (0.0, 10.0], else keeps current value.

`gdelay_feed` Set feedback amount.

Range: 0.0 to 1.0

0.0 = single echo

1.0 = infinite regeneration

Clamped to bounds.

Feedback applied after tone filter in feedback loop (DD-4 and Bencina modes only; inactive in Stut mode).

Signal inlet 11: updates if value in range (0.0, 1.0], else keeps current value.

`gdelay_tone` Set tone character via one-pole IIR low-pass filter in feedback loop (DD-4 and Bencina modes only; inactive in Stut mode).

Range: 0.0 to 1.0

0.0 = dark, heavily filtered (analog-style decay)

1.0 = bright, no filtering (digital-style)

Clamped to bounds.

Filter coefficient applied directly:  $\text{filtered} = \text{tone} \times \text{delayed} + (1 - \text{tone}) \times \text{lpf\_state}$

Higher values pass more high frequencies. Lower values create darker, warmer repeats.

Signal inlet 12: In DD-4/Bencina modes, updates feedback if value in range (0.0, 1.0]. In Stut mode (delay\_mode 2), controls stut\_reduction (gain decay per repeat, range 0.0-1.0).

gdelay\_mix Set dry/wet mix.

Range: 0.0 to 1.0

0.0 = 100% dry (grain output only, no delay)

1.0 = 100% wet (delayed signal only)

Clamped to bounds.

Output calculation:  $\text{output} = \text{dry} \times (1 - \text{mix}) + \text{wet} \times \text{mix}$

Signal inlet 13: In DD-4/Bencina modes, updates tone if value in range (0.0, 1.0]. In Stut mode (delay\_mode 2), controls stut\_spacing (repetition spacing in ms, range 1.0-5000.0).

gdelay\_clear Clear delay buffers and filterstate.

Zeros buffer\_left, buffer\_right, lpf\_state\_left, lpf\_state\_right.

Immediate silence in feedback loop.

Delay Modes

delay\_mode <0|1|2> Select delay algorithm.

DD-4 Mode (delay\_mode 0)

Default mode. Classic analog-style delay with a single read head at a fixed distance behind the write head. Feedback and tone filter operate in the feedback loop.

Architecture: input + feedback is written to the buffer. A single read head reads at write\_pos minus delay\_time with linear interpolation. The delayed signal passes through a one-pole IIR low-pass filter (the tone control), then feedback is calculated from the filtered output.

Active inlets: all four (time, feedback, tone, mix).

Delay time is smoothed internally (coefficient 0.001, approximately 20ms transition at 48kHz) to prevent clicks during time changes.

Bencina Mode (delay\_mode 1)

Pitch-preserving granular delay based on the Ross Bencina architecture. Instead of a single read head, spawns up to 32 micro-grains inside the delay buffer. Each grain stores a relative offset from the write head (a “moving tap”), so the grain follows the write head at a fixed distance. This is what preserves pitch — the grain envelope re-synthesizes the delayed audio independently of delay time changes.

Architecture: grains are triggered continuously at a fixed interval (`bencina_iot`). Each grain reads from `write_pos` minus its stored offset, applies the main envelope shape as a window, and uses constant-power panning. The sum of all active grains passes through the tone filter, and the filtered output is fed back into the buffer before the write head advances. This true feedback creates recursive granulation — delayed grains get re-granulated — producing shimmer and evolving textures at higher feedback values.

Active inlets: all four (time, feedback, tone, mix). Time controls the distance of the moving tap from the write head. Feedback and tone operate in the grain output path.

`bencina_iot` Spacing between grain triggers (milliseconds). Controls grain density.

Range: 1.0 to 1000.0 ms. Default: 50.0 ms (20 grains/sec).

Lower values = denser cloud, more overlap between grains. For rhythmic effects, set to a musical subdivision (e.g. 125ms = 1/8 note at 120 BPM).

`bencina_grainsize` Individual grain length (seconds).

Range: 0.001 to 2.0 seconds. Default: 0.1 seconds (100ms).

Longer grains = smoother, more blurred output. Shorter grains = more granular texture. Applied to future grains only (existing grains keep their size).

`bencina_wrap` <0|1> Buffer wrap mode.

0 = global (default): grains can read from any part of the buffer, creating layered textures from across the recording. 1 = splice: grains wrap within current splice boundaries, preventing cross-splice bleed.

Applied to all existing and future grains.

`bencina_clear` Clear all active `bencina` grains. Immediate silence. Resets active grain count to 0.

Stut Mode (`delay_mode` 2)

Quantized rhythmic delay. Unlike DD-4 and Bencina which run continuously, Stut is event-triggered via the `stut` message. When triggered, it captures the current buffer write position and schedules `N` repetitions at fixed time intervals, each with exponentially decaying gain.

Architecture: the `stut` message captures an absolute buffer position at the moment of triggering. It then schedules up to 16 grains, each with a countdown timer ( $i \times \text{spacing}$ ) and a gain value ( $\text{gain\_reduction}^i$ ). During processing, each grain counts down; when its timer reaches zero it emits a single-sample pulse from the captured position, scaled by its gain. This produces discrete rhythmic echoes of the captured moment.

Active inlets: time, mix, and — in Stut mode — inlets 12 and 13 are repurposed. Inlet 12 controls `stut_reduction` (gain decay per repeat, 0.0-1.0) instead of feedback. Inlet 13 controls `stut_spacing` (repetition spacing in ms, 1.0-5000.0) instead of tone. The stut grains read directly from captured buffer

positions with per-repetition gain decay — there is no feedback loop or tone filter in the stut signal path. The mix parameter controls dry/wet blend as usual. Modulation ranges (gdelay\_feed range, gdelay\_tone range) also route to stut\_reduction and stut\_spacing respectively when in Stut mode.

stut Trigger a stut sequence. Captures the current splice boundaries and write position. Schedules repetitions immediately. If delay quantization is active (delay\_quant > 0, BPM running), uses the quantized grid spacing instead of stut\_spacing.

stut\_reps Number of repetitions per trigger.

Range: 1 to 16. Default: 4.

stut\_reduction Gain reduction factor per repeat. Each repetition's gain = reduction ^ repetition\_number.

Range: 0.0 to 1.0. Default: 0.5.

0.0 = maximum reduction (only the first repeat is audible). 0.5 = each repeat is half the volume of the previous. 1.0 = no reduction (all repeats at full volume).

Example with 4 reps, reduction 0.5: gains are [1.0, 0.5, 0.25, 0.125].

stut\_spacing Base spacing between repetitions (milliseconds).

Range: 1.0 to 5000.0 ms. Default: 62.5 ms (1/16 note at 120 BPM).

Overridden by delay quantization grid when active (see below).

## Stut and Delay Quantization

The stut trigger integrates with the delay quantization system. When all three conditions are met — delay\_quant amount > 0, BPM > 1.0, and clock running — the stut trigger uses delay\_quant\_grid\_ms as the spacing between repetitions instead of stut\_spacing. This locks stut repeats to the tempo grid.

Example: at 120 BPM with delay\_quantize 16 and delay\_quant 1.0, stut repeats are spaced at 125ms (1/16 note). Changing delay\_quantize to 8 spaces them at 250ms (1/8 note).

Note: delay quantization also affects DD-4 delay time (blending toward the grid), but does not affect Bencina grain spacing. Use bencina\_iot directly for Bencina timing.

## Per-Mode Parameter Summary

|             |                   |            |             |              |            |            |              |        |             |             |                |
|-------------|-------------------|------------|-------------|--------------|------------|------------|--------------|--------|-------------|-------------|----------------|
| DD-4        | Bencina           | Stut       | gdelay_time | Active       | Active     | Inactive*  | gdelay_feed  | Active | Active      | →           | stut_reduction |
| gdelay_tone | Active            | Active     | →           | stut_spacing | gdelay_mix | Active     | Active       | Active | delay_quant | Blends time | No effect      |
| Overrides   | spacing           | Triggering | Continuous  | Continuous   | Event      | (stut msg) | Read pattern | Fixed  | offset      |             |                |
| Moving tap  | Captured position | Max voices | 1           | 32           | 16         |            |              |        |             |             |                |

- Stut writes input to the shared buffer using the write head, but stut grains read from captured absolute positions. The delay\_time parameter does not affect stut grain playback positions.

All three modes share the circular delay buffer and write head. Mode-specific parameters only affect their respective mode. Switching modes via `delay_mode` takes effect immediately.



## FOG (SPECTRAL EFFECT)

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FFT-based spectral processing applied to the grain output. Combines horizontal smear (frequency-axis magnitude averaging) and vertical specmagfilter (time-axis resonant filtering of magnitude and phase) to create spectral fog — timbral sustain, blurring, and ghostly persistence.

Signal chain: FFT → polar conversion → smear → specmagfilter → IFFT

Fog is a monitoring effect — it is not recorded into the reel. Applied after delay and before distortion in the effects chain.

Signal inlet 15: fog mix (0.0-1.0). 0.0 = dry bypass (no FFT processing). 1.0 = full wet (100% spectral effect). Equal-power crossfade between dry and wet.

### Smear (Horizontal)

Frequency-axis magnitude averaging. Blurs the spectral content across neighboring bins, creating a diffuse, hazy timbre.

`fog_smear_bins` Number of neighbor bins to average on each side.

Range: 0 to 32. Default: 8.

Higher values = wider blur, more diffuse timbre. 0 = no smearing.

`fog_smear_enable` <0|1> Enable/disable smear processing.

`fog_smear_onset_curve` <0|1|2> How smear amount scales with fog mix. 0 = linear, 1 = exponential ( $x^2$ , delayed onset), 2 = logarithmic (sqrt, early onset). Default: 2.

`fog_smear_onset_amount` <0-1> Maximum smear depth at mix=1.0. Default: 1.0.

### SpecMagFilter (Vertical)

Time-axis resonant filtering. Lowpass filters magnitude and phase per bin across consecutive FFT frames, creating temporal persistence — spectral content decays slowly rather than updating instantly.

`fog_mag_cutoff` Magnitude filter cutoff frequency. Controls how quickly magnitudes track changes.

Range: 0.1 to 20.0 Hz (relative to FFT hop rate). Default: 2.0 Hz.

Lower values = slower tracking, longer spectral sustain. Higher values = faster response.

`fog_mag_resonance` Magnitude filter resonance (Q factor). Adds a peak at the cutoff frequency, emphasizing the rate of spectral change.

Range: 0.1 to 10.0. Default: 1.0.

Higher values create a more pronounced, ringy spectral persistence. Soft-limited via tanh to prevent magnitude explosion.

`fog_phase_cutoff` Phase lowpass cutoff. Controls smoothing of phase evolution between frames.

Range: 0.1 to 20.0 Hz. Default: 2.0 Hz.

Lower values freeze phase relationships, creating a static spectral snapshot.

`fog_specmagfilter_enable` <0|1> Enable/disable temporal filtering.

`fog_specmagfilter_onset_curve` <0|1|2> How filter depth scales with fog mix. 0 = linear, 1 = exponential, 2 = logarithmic (default).

`fog_specmagfilter_onset_amount` <0-1> Maximum filter depth at mix=1.0. Default: 1.0.

#### Stereo Filter Mode

`fog_stereo_filter_mode` <0|1> Select stereo processing strategy.

0 (default): Shared state. Left and right channels share the same per-bin filter state arrays. Creates a diffuse, mono-ish spectral character — both channels converge toward the same spectral evolution.

1: Independent. Each channel has its own per-bin state arrays. Left and right channels evolve independently, preserving true stereo spectral separation.

#### Per-Grain Fog Mode

`fog_position` <0|1> Select fog routing.

0: Per-grain. Each grain is assigned to one of N independent fog slots via round-robin at trigger time. Grains accumulate into per-slot buffers, then each slot runs its own FFT pipeline. Different grains develop independent spectral evolution — some fogging faster, some slower — creating richer, more complex textures than global processing.

1 (default): Post-mix. Single fog instance processes the mixed grain output. Existing behavior.

Pool size is configurable via `ligase.conf` (`fog_pool_size`, 1-8 slots, default 4). Each slot costs ~76KB.

#### Configuration (`ligase.conf`)

`fft_size` = 1024 FFT window size: 512, 1024 (default), or 2048. 512 = lower latency, less frequency resolution. 2048 = highest resolution, higher latency.

`overlap_factor` = 4 Overlap factor: 2, 4 (default), or 8. Higher = smoother temporal evolution, more CPU.

`fog_pool_size` = 4 Per-grain fog slots: 1-8, default 4. ~76KB per slot.

# DISTORTION

---

Multi-mode waveshaping with pre/post filters and oversampling. Two position modes: per-grain (experimental) or post-mix (default). In post-mix mode, distortion is applied only AFTER recording as a monitoring effect.

Control

distortion\_enable 0|1 Enable/disable distortion processing.

0 = bypass (passthrough)

1 = enabled

Signal inlet 14: updates if value in range (0.0, 1.0], else keeps current value.

distortion Set distortion drive amount.

Range: 0.0 to 1.0

Maps to current\_drive via lookup curve.

0.0 = clean (minimal drive)

1.0 = saturated (maximum drive)

distortion\_position 0|1 Set distortion placement in signal chain.

Mode 0: Per-grain (experimental)

Applied to each grain sample during synthesis, before envelope. No oversampling. Uses grain\_distortion\_process\_sample().

Signal chain: read → distort → envelope → pan → mix

Distortion IS recorded in this mode (affects grain output before recording).

Mode 1: Post-mix (default)

Applied to final output AFTER all recording operations.

Uses oversampling. Uses grain\_distortion\_process\_block().

Signal chain: grains → delay → [RECORDING] → fog → distortion → Moog → dac~

Distortion NOT recorded (monitoring/output only).

This placement prevents distortion artifacts from accumulating in overdub recordings.

Default: 1 (post-mix)

distortion\_oversample\_factor 1|2|4|8 Set oversampling multiplier for post-mix mode. Valid values: 1, 2, 4, 8 ?

Posts error if invalid value provided.

Higher values reduce aliasing at cost of CPU.

4x provides good quality/performance balance.

Per-grain mode ignores this setting (no oversampling).

Default: 4

### Waveshaper Modes

dist\_waveshaper\_mode 0|1|2|3|4 Select waveshaping algorithm.

Mode 0: tanh (default)

Pure hyperbolic tangent.

Aggressive odd harmonics. Warm, tube-like saturation.

Mode 1: arctan

Pure arctangent.

Smooth odd harmonics. Gentler than tanh.

Mode 2: asymmetric

Different drive for positive/negative cycles.

Generates even harmonics.

Uses drive\_pos and drive\_neg parameters.

Mode 3: blend

Morph between tanh and arctan.

Uses dist\_curve\_blend parameter (0.0 = tanh, 1.0 = arctan).

Mode 4: polynomial

User-defined polynomial:  $\text{output} = c1 \times x + c2 \times x^2 + c3 \times x^3$

Uses poly\_c1, poly\_c2, poly\_c3 coefficients.

Precise harmonic control.

Default: 0 (tanh)

dist\_curve\_blend Blend amount for Mode 3.

Range: 0.0 to 1.0

0.0 = full tanh (aggressive)

1.0 = full arctan (smooth)

dist\_drive\_pos Positive cycle drive for Mode2. Range: 1.0 to 20.0

Higher values = more saturation on positive peaks.

dist\_drive\_neg Negative cycle drive for Mode2.

Range: 1.0 to 20.0

Higher values = more saturation on negative peaks.

dist\_poly\_c1 Linear coefficient for Mode 4.

Range: -10.0 to 10.0

Controls fundamental and odd harmonics.

Default: 1.0 (linear passthrough)

dist\_poly\_c2 Quadratic coefficient for Mode 4.

Range: -10.0 to 10.0

Controls even harmonics (asymmetry).

dist\_poly\_c3 Cubic coefficient for Mode 4.

Range: -10.0 to 10.0

Controls additional odd harmonics.

Pre-emphasis / De-emphasis (Inner Filters)

Frequency-dependent pre-distortion boost with matched post-distortion cut. Affects harmonic content without changing overall tonal balance.

dist\_emphasis\_mode 0|1 Set pre-emphasis filter type.

0 = HP (high-pass): boost highs before distortion → brighter saturation

1 = LP (low-pass): boost lows before distortion → darker saturation

Argument  $\geq 0.5$  selects LP,  $< 0.5$  selects HP.

Default: 0 (HP mode)

dist\_emphasis\_freq Set emphasis corner frequency. Range: 100 to 5000 Hz

Higher freq = more extreme emphasis effect.

dist\_pregain Set pre-distortion gain stage.

Range: 0.1 to 10.0

Applied before emphasis filters and waveshaper.

1.0 = unity  
1.0 = drive harder

$< 1.0$  = soften

Outer Filters

Applied outside emphasis/waveshaper core. Shape frequency response without affecting emphasis balance.

distortion\_pre\_hp\_freq Pre-tanh highpass frequency.

Range: 30 to 500 Hz

1-pole filter (6dB/octave).

Removes DC and low mud before saturation.

distortion\_pre\_hp\_mix Pre-tanh highpass dry/wet.

Range: 0.0 to 1.0

0.0 = full passthrough (no filtering)

1.0 = full filtered

distortion\_post\_lp\_freq Post-tanh lowpass frequency.

Range: 2400 to 10000 Hz

2-pole Butterworth (12dB/octave).

Tames high-frequency harmonics after saturation.

distortion\_post\_lp\_mix Post-tanh lowpass dry/wet.

Range: 0.0 to 1.0

0.0 = full bright (no filtering)

1.0 = full dark

distortion\_notch\_freq Notch filter centerfrequency.

Range: 20 Hz to (sample\_rate/2 - 100) Hz

Bandstop filter for targeted frequency removal.

distortion\_notch\_bw Notch filter bandwidth.

Range: 10 to 5000 Hz

Narrower = more surgical cut.

distortion\_notch\_mix Notch filter dry/wet.

Range: 0.0 to 1.0

0.0 = inactive (no notching)

1.0 = full notch applied

Default: 0.0 (inactive)

# MOOG LADDER FILTER

---

Classic 4-pole (24dB/octave) Moog ladder lowpass filter. Four cascaded one-pole stages with resonance feedback. Filter is applied only AFTER recording as a monitoring effect.

## Parameters

moog\_cutoff Set filter cutoff frequency.

Range: 20 to 20000 Hz

Clamped to Nyquist frequency ( $\text{sample\_rate} / 2$ ).

Formula:  $f = 2 \times \sin(\pi \times f_c / f_s)$

Cutoff coefficient clamped to [0.0, 1.0] for stability.

Signal inlet 16: updates if value in range [20.0, 20000.0], else keeps current value.

moog\_resonance Set resonance feedback amount.

Range: 0.0 to 4.0

0.0 = no resonance (flat lowpass)

3.5+ = self-oscillation (pitched sine tone at cutoff frequency)

4.0 = maximum feedback

Feedback formula:  $fb = \text{resonance} \times (1.0 + 0.5 \times f^5)$

Nonlinear compensation prevents resonance drop at high cutoff frequencies.

Signal inlet 16: updates if value in range [0.0, 4.0], else keeps current value.

moog\_mix Set dry/wet mix.

Range: 0.0 to 1.0

0.0 = 100% dry (bypass, no filtering)

1.0 = 100% wet (fully filtered)

Output:  $\text{output} = \text{dry} \times (1 - \text{mix}) + \text{wet} \times \text{mix}$

Signal inlet 17: updates if value in range [0.0, 1.0], else keeps current value.



moog\_enable 0|1 Enable/disable filter processing.

0 = bypass (passthrough)

1 = enabled

When disabled or mix = 0.0, filter bypasses without processing.

# PARAMETER RANGES & MODULATION

---

Per-grain parameter variation system. Each parameter can vary within a specified range driven by one of nine generator types (rand, Perlin 1D/2D, Lorenz, N-body, sphere, saw, sine, square — four instances each). ~37 parameters across grain synthesis, timing/playback, delay, distortion, fog, and the modulation outlets are modulatable.

## Modulation Sources

Five generator types, each with 4 independent instances (1-4):

- **RAND\_TYPE\_RAND**

Basic pseudo-random. Seeded per instance. Uniform distribution [0.0, 1.0]. Fast, uncorrelated white noise variation.

- **RAND\_TYPE\_PERLIN\_1D**

1D Perlin noise. Smooth continuous modulation. Coordinate advances by  $\text{iot} \times \text{noise\_frequency\_scale}$  per grain. Decorrelated via instance offsets: [0, 1000, 2003, 3001]. Returns normalized [0.0, 1.0].

- **RAND\_TYPE\_PERLIN\_2D**

2D Perlin noise. X-coordinate advances per grain, Y-coordinate = base parameter value.

Coupling between parameter value and modulation depth. Decorrelated via instance offsets: [0, 5000, 10007, 15013]. Returns normalized [0.0, 1.0].

- **RAND\_TYPE\_LORENZ**

Lorenz attractor. Bounded chaotic motion. Evolves via update iterations:  $\text{int}(\text{iot} \times 100 \times \text{noise\_frequency\_scale})$ . Capped 1-50 iterations per grain for stability. Output axis rotates per instance: 0=X, 1=Y, 2=Z (mod 3). Different initial positions per instance for decorrelation. Returns normalized [0.0, 1.0].

- **RAND\_TYPE\_NBODY**

N-body gravitational simulation (3-body problem). Bounded chaotic orbital motion.

Evolves via update iterations like Lorenz. Output mode selectable per instance (0-10):

Body positions, velocities, distances, angular momentum, total energy. Returns normalized [0.0, 1.0].

- **RAND\_TYPE\_SPHERE**

3D sphere physics simulation (STK-based). A sphere bounces within a bounded space with configurable damping and elasticity. Each instance can be “kicked” with a velocity impulse. Output mode selectable per instance (0-6): 0=Position X, 1=Position Y, 2=Position Z, 3=Velocity X, 4=Velocity Y, 5=Velocity Z, 6=Velocity Magnitude. Returns normalized [0.0, 1.0].

sphere\_kick Apply velocity impulse to sphere instance. Creates sudden, decaying motion patterns.

sphere\_damping <0-1> Set damping coefficient. Higher values = faster energy decay.

sphere\_elasticity <0-1> Set bounce elasticity. 1.0 = perfect elastic bounce. Lower values absorb energy on wall contact.

sphere\_reset Reset sphere to initial position and zero velocity.

sphere\_mode <0-6> Set output mode. 0-2 = position axes, 3-5 = velocity axes, 6 = velocity magnitude.

- RAND\_TYPE\_SAW

Sawtooth wave LFO. Unipolar ramp from 0.0 to 1.0. Phase advances by  $\text{iot} \times \text{noise\_frequency\_scale}$  per grain trigger. Creates predictable linear sweep patterns for modulation.

- RAND\_TYPE\_SINE

Sine wave LFO. Unipolar 0.0 to 1.0 (mapped from bipolar sine). Phase advances by  $\text{iot} \times \text{noise\_frequency\_scale}$  per grain trigger. Creates smooth, predictable oscillating modulation.

- RAND\_TYPE\_SQUARE

Square wave LFO. Bipolar switching: 0.0 below 50% duty cycle, 1.0 above. Phase advances by  $\text{iot} \times \text{noise\_frequency\_scale}$  per grain trigger. Creates alternating on/off modulation patterns.

Modulatable Parameters Include:

- Grain synthesis (sampled per grain trigger):

speed, grainsize, grainstart, amplitude, pan, maxgrains, env\_skew

- Timing / playback (sampled per DSP block):

iot (interonset time), scanrate, organize (splice select), sos (sound-on-sound)

- Effects (sampled per DSP block):

gdelay (delay time), gdelay\_feed, gdelay\_tone, gdelay\_mix, moog\_cutoff, moog\_resonance, moog\_mix

- Distortion:

distortion (intensity), dist\_emphasis\_freq, dist\_pregain, dist\_curve\_blend, dist\_drive\_pos, dist\_drive\_neg, dist\_poly\_c1, dist\_poly\_c2, dist\_poly\_c3

- Fog (sampled per DSP block):

fog\_mix, fog\_smear\_bins, fog\_smear\_onset, fog\_mag\_cutoff, fog\_mag\_resonance, fog\_phase\_cutoff, fog\_smf\_onset

- Stut (sampled per DSP block):

stut\_reps

- Bencina/PPG (sampled per DSP block):

bencina\_iot, bencina\_grainsize

- Modulation outlets (the four send outlets are themselves modulatable targets):

modout1, modout2, modout3, modout4

Messages

- param\_range [max] Set parameter modulation range.
  - Two-argument form (single value): Disables range, sets fixed value. Example: param\_range speed 1.0→ speed fixed at 1.0
  - Three-argument form (range): Enables modulation between min and max. Example: param\_range speed 0.5 2.0→ speed varies 0.5-2.0
- rand\_type [param\_name] Set modulation source for parameter(s).
  - type\_instance format: {type}\_{N} where N = 1-4
  - Types: rand\_1...rand\_4, perlin\_1d\_1...perlin\_1d\_4, perlin\_2d\_1...perlin\_2d\_4, lorenz\_1...lorenz\_4, nbody\_1...nbody\_4, sphere\_1...sphere\_4, saw\_1...saw\_4, sine\_1...sine\_4, square\_1...square\_4
  - One-argument form: Sets all parameters to specified source. Example:

rand\_type perlin\_1d\_2

- Two-argument form: Sets single parameter to specified source. Example:

rand\_type lorenz\_3 amplitude

- noise\_freq Set frequency scale for all 4 modulation instances.
  - Range: > 0.0 (1.0 = normal speed)

- Affects Perlin coordinate advancement and Lorenz/N-body iteration count.

- noise\_freq\_1
- noise\_freq\_2
- noise\_freq\_3
- noise\_freq\_4

Set frequency scale for specific instance (1-4). Independent control of each modulation instance speed.

Lock

Sends current modulated values on outlet 9 then stops modulation.

param\_lock ...

Example:

param\_range pan 0.2 0.8

rand\_type perlin\_1d\_1 pan

param\_lock pan → “ligase~: pan locked at 0.547”

param\_lock pan amplitude grainsize → locks all three at current values

Disables range modulation only.

Returns to inlet or last message value.

Without feedback routing:

→ Just stops modulation

→ Returns to inlet/message control

→ Values go to outlet 9 but aren’t applied

With feedback routing (outlet 9 → inlet 1):

→ Stops modulation

→ Captures and applies current modulated values

→ “Locks” parameters at modulated state

N-Body Control

N-body instances have additional physics parameters (1-4):

- `nbody_epsilon` Set softening parameter (0.001-0.1). Prevents singularities during close approaches.
- `nbody_damping` Set velocity damping (0.0-0.1). Higher values = faster energy loss, more stable orbits.
- `nbody_pump` Set energy pump interval (velocity increment, iteration interval). Adds velocity periodically to sustain motion
- `nbody_G` Set gravitational constant(0.1-10.0).
- `nbody_mode` Set output extraction mode (0-10).

11 output modes per instance:

- 0: Body 0 X position (instance 1 default)
- 1: Body 1 Y position (instance 2 default)
- 2: Body 2 X position (instance 3 default)
- 3: Distance body 0-1 (instance 4 default)
- 4: Velocity magnitude body 0
- 5: Velocity magnitude body 1
- 6: Velocity magnitude body 2
- 7: Distance body 0-2
- 8: Distance body 1-2
- 9: Angular momentum magnitude
- 10: Total system energy

Auto-normalizing bounds track min/max values.

Sampling Timing

- Per grain trigger: speed, grainsize, grainstart, amplitude,pan, and maxgrains
- Per DSP block (smooth variation): `iot`, `gdelay`, `moog_cutoff`,`moog_resonance`, `moog_mix`.

Modulation Mapping

Output value:  $\text{min} + \text{random\_value} \times (\text{max} - \text{min})$ . Linear mapping using normalized random\_value [0.0, 1.0] from the generator.

### Generator Outlet Sends

The four modulation outlets (modout1–modout4) are first-class modulatable parameters. Configure them with the same unified messages used for any parameter, addressing them by name (modout1... modout4); there are no separate modout\_source/modout\_range commands:

rand\_type modoutN - assign a generator (any of the nine types, instance 1-4) to outlet N

param\_range modoutN - set the outlet's output range

param\_slew / param\_invert modoutN - smooth / invert the outlet, as for any parameter

### Per-Outlet Base Value

Set PERLIN\_2D Y-coordinate for parameter decorrelation.

param\_base\_value

Example:

rand\_type perlin\_2d\_1 modout1

rand\_type perlin\_2d\_1 modout2

param\_base\_value modout1 0.2

param\_base\_value modout2 0.8

Result:

modout1 and modout2 use same instance but different Y-coordinates, producing decorrelated patterns.

Notes:

Default: 0.5 for all parameters

Only affects PERLIN\_2D (ignored for other generator types)

Enables spatial decorrelation without using multiple instances

### Parameter Slewing

Apply exponential smoothing to modulation output.

param\_slew

Example:

rand\_type rand\_1 modout1

param\_range modout1 0 1

param\_slew modout1 0.9 → Very smooth, slow tracking

param\_slew modout1 0.5 → Moderate smoothing

param\_slew modout1 0.0 → Instant (no smoothing)

Range:

0.0: Instant response (disabled)

0.5: Moderate smoothing

0.99: Very slow tracking

1.0: Frozen (no updates)

Notes:

Algorithm:  $\text{smoothed} = \text{smoothed} \times \text{slew} + \text{new} \times (1 - \text{slew})$

Works with all generator types

Generator Reset

Reset Perlin or Lorenz generators to initial state.

perlin\_reset

lorenz\_reset

Example:

perlin\_reset 1 → Resets Perlin 1D/2D coordinates to 0

lorenz\_reset 2 → Resets Lorenz attractor to initial position

nbody\_reset 3 → Resets N-body system (existing feature)

Perlin Reset:

Resets noise\_1d\_coord and noise\_2d\_coord\_x to 0.0

Useful for synchronizing modulation patterns

Lorenz Reset:

Returns x, y, z to initial values (x0, y0, z0)

Each instance has different starting position:



Instance 1: (0.1, 0.0, 0.0)

Instance 2: (5.0, 10.0, 20.0)

Instance 3: (-3.0, -5.0, 10.0)

Instance 4: (8.0, 15.0, 30.0)

N-body Reset:

Resets to hierarchical initial configuration

Each instance has different mass distribution

Notes:

Instance range: 1-4

Does not reset frequency scaling (perlin\_frequency)

Does not reset N-body physics parameters (epsilon, damping, etc.)

Defaults

All parameter ranges initialized to: min: 0.0, max: 1.0, rand\_type: RAND\_TYPE\_RAND (instance 1), enabled: 0 (disabled). All noise\_frequency\_scale defaults to 1.0.

## QUERY STATE

---

Export and inspect ligase~ state via outlet 9.

Useful for creating presets or routing values to other instances:

query → Output single parameter value

get\_params → Output all 20 parameter values

Debugging and inspection:

get\_inlets → Print inlet guide to console

get\_ranges → Output active param\_range settings

get\_generators → Output active rand\_type assignments

get\_state → Output everything (params + ranges + generators)

query bpm → returns calculated bpm

get\_inlets

Print inlet documentation to console (no outlet).

Output: Console only

Shows current values and message syntax

query

Query single parameter value.

query

20 queryable parameters:

speed, grainsize, grainstart, organize, scanrate, sos, iot, maxgrains, gdelay, gdelay\_feed, gdelay\_tone, gdelay\_mix, distortion, moog\_cutoff, moog\_resonance, moog\_mix, midi, env\_skew, amplitude, pan

Per-block parameters - store modulated values when param\_range.enabled: gdelay\_time, gdelay\_feedback, gdelay\_tone, gdelay\_mix moog\_cutoff, moog\_resonance, moog\_mix organize, sos, env\_skew, iot

Per-grain parameters - Sample once per block for query when modulation enabled:

speed, grainsize, grainstart, amplitude, pan, maxgrains, distortion

query and get\_params return:

Inlet/ last message value when modulation is disabled

Current modulated value when modulation is active

Locked value when param\_lock has been applied

Unconnected inlets preserve stored values if headless mode is enabled

get\_params

Export all 20 parameter values.

Fixed order (speed → grainsize → ... → pan)

get\_ranges

Export active parameter range configurations.

Output format:

param\_range

Only outputs enabled ranges

Messages can be sent back to restore range configuration

get\_generators

Export active generator assignments.

Output format:

rand\_type \_

Generator types: rand, perlin\_1d, perlin\_2d, lorenz, nbody

Instance numbers 1-4

Only outputs modout1-4 (scheduler generators not accessible)

Does NOT export perlin\_frequency or nbody settings

get\_state

Export complete state (all three subsystems).

Output order: 1. All parameter values (20 messages) 2. All enabled ranges (variable) 3. All generator assignments (variable)

Calls `get_params` + `get_ranges` + `get_generators`

All messages are routable lists

Limitations

No quantization state - timesig/quant settings not exported